

Assignment 2: Motherboard and CPU Theory

Task 1: Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).

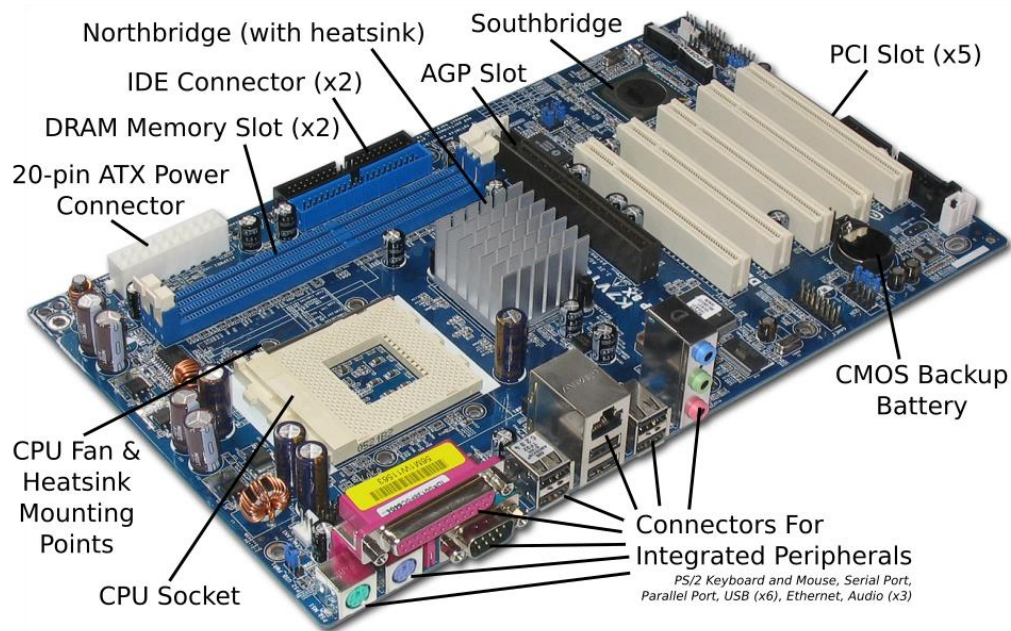


Fig. 1. ATX Motherboard with ports

Task 2: Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

Type	CPU Model
LGA1151	Intel Core i7-9700K
AM4	AMD Ryzen 5 5600X
LGA1200	Intel Core i5-10400

Task 3: Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

CPU: The CPU is the main processor of the computer. It executes instructions, performs calculations, and controls the operation of programs. When you open Spotify, the CPU processes the application's code, loads the user interface, and manages music playback.

Chip Set: The chipset is a group of controller chips on the motherboard that manages communication between the CPU and other hardware such as RAM, storage drives, USB devices, and audio components. It ensures that data flows efficiently between all connected devices.

Opening Spotify: When you open Spotify, the CPU begins executing the program's instructions. The chipset helps transfer data between the SSD (where Spotify is stored), the RAM (where it loads), the CPU (which processes it), the network adapter (for internet streaming), and the audio hardware (to play music). Together, they allow Spotify to start quickly, stream songs, and output sound through the speakers or headphones.

Task 5: Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labelling each one.

